Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/30/2005
PI Name:	Graves, Lee M Ph.D.		
Project Title:	Application Of Proteomics To Study Skeletal Muscle Atrophy Resulting From Exposure To Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	None		
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Zip Code:	27599-7365	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/15/2002	**End Date:**	11/14/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1599		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We propose to use modern techniques of mass spectrometry and "proteomics" to analyze both global and specific changes in protein expression during muscle atrophy. Analogous to "gene chips," this approach will allow us to identify many of the key enzymes affected by muscle wasting. In addition, we will focus on characterizing the change in tyrosine phosphorylation of key intracellular signaling proteins and identifying the target signaling molecules influenced during this process. Finally the ability of growth stimulatory factors or hormones to reverse this process of muscle atrophy will be examined. These studies will thus be instrumental in identifying the changes that occur in skeletal muscle upon wasting and will provide fundamental information required for the design of therapeutic agents useful against this condition. To determine how skeletal muscle atrophy influences the development of insulin resistance through changes in the regulation of protein kinase signaling. Important metabolic change during atrophy that may influence the function of skeletal muscle in astronauts exposed to low gravity conditions.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Not Available
Task Progress:	A large study was conducted comparing the biological effects of low gravity (hindlimb suspension) to those induced by high gravity (2G) by centrifugation. Soleus and gastrocnemius muscle samples obtained from rats were examined for changes in expression and phosphorylation of a number of important signal transduction kinases. In addition, the phosphorylation status of proteins and receptors involved in the uptake of glucose and cellular responses to insulin were determined. The results of these studies demonstrated the time-dependent increase in phosphorylation of the c-jun (JNK) Map kinases in responses to low gravity conditions. Similarly an increase in p38 MAP kinases and the energy sensing AMP-activated (AMPK) were detected duing exposure to low gravity conditions. Corresponding changes in phosphorylation of the Insulin receptor substrate (IRS-1) was observed that correlated with activation of JNK in these samples. Interestingly, these changes were specific to anti-gravity muscles (soleus) and were not observed in the high gravity samples. In addition, important difference were observed in the expression of the Glucose transport proteins (GLUT1/4) demonstratign that low gravity conditions disrupt the normal metabolic repsonses to insulin-mediated glucose uptake and metabolism. These studies were accepted for publication in 2005 (Hilder et. al. 2005). In summary our studies accomplished the following: Through the application of proteomics and other methods we determined that a number of important cellular signal transduction kinases were influence by exposure of animals to low gravity conditions. The most important of these was JNK, which is a negative regulator of insulin receptor signaling, was markedly activated by hind limb suspension and not high gravity exposure. The implication of these studies will be that low gravity conditions lead to impaired insulin receptor signaling and disposition which may influence the process of atrophy and/or decreased muscle function during space travel. Our studies also demonstrated unexpected effects of these treatments on glucose transporter function and suggest that a change in energy metabolism is indicated by activation of the AMP-dependent protein kinase. This too is expected to contribute to the remodeling of the physiological properties of skeletal muscle during muscle disuse atrophy. Finally in a related study, we demonstrated that during muscle death a key enzyme in glycogen metabolism was proteolyzed by a specific class of proteases know as caspases. This cleavage of this enzyme resulted in activation and changes in glycogen and glucose metabolism in wasting muscle. These studies resulted in 3 peer reviewed publications.
Bibliography Type:	Description: (Last Updated: 10/21/2009)
Articles in Peer-reviewed Journals	Hilder TL, Baer LA, Fuller PM, Fuller CA, Grindeland RE, Wade CE, Graves LM. " Insulin-independent pathways mediating glucose uptake in hindlimb-suspended skeletal muscle." J Appl Physiol. 2005 Dec;99(6):2181-8. Epub 2005 Aug 11. PMID: 16099889 , Dec-2005
Articles in Peer-reviewed Journals	Hilder TL, Carlson GM, Haystead TA, Krebs EG, Graves LM. "Caspase-3 dependent cleavage and activation of skeletal muscle phosphorylase b kinase." Mol Cell Biochem. 2005 Jul;275(1-2):233-42. PubMed PMID: 16335803 , Jul-2005